Western Ghats Deforestation & Land Cover Change Detection

Region: **Wayanad** | Years: 2025 → 2026

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Alert Level: Low Risk

Forest Cover Summary

Metric	Value
Forest Cover (Year 1)	56.00%
Forest Cover (Year 2)	55.68%
Forest Loss	0.31%
Forest Gain	0.00%
Net Forest Change	-0.31%

Land Cover Percentages

Class	Year 2025 %	Year 2026 %	Change %
Forest	56.00%	55.68%	-0.31%
Agriculture	0.04%	0.04%	-0.01%
Urban	16.86%	17.76%	+0.90%
Water	26.50%	22.24%	-4.26%
Barren	0.59%	4.28%	+3.69%

Key Land Cover Transitions

Transition	Area (%)
Forest → Urban	2.063%
Forest → Agriculture	0.017%
Forest → Barren	0.011%
Agriculture → Urban	0.001%

Recommendations

• Urban expansion detected near forest boundary.

• Net forest gain detected — positive trend, continue monitoring.

Academic Analysis & Ecological Justification

1. Reasoning for Land Cover Shifts

A significant reduction of 0.31% in dense forest cover indicates acute anthropogenic pressures. Studies such as those by the Indian Institute of Science (IISc) on the Western Ghats highlight that unbridled agricultural expansion and infrastructure projects are prime catalysts for this deforestation. The 0.90% expansion in built-up urban areas is a direct consequence of rapid demographic shifts and infrastructure development. Recent environmental appraisals emphasize that unregulated urbanization in the eco-sensitive zones of the Western Ghats severely fragments primary habitats. A concerning reduction of 4.26% in water bodies correlates with reduced groundwater recharge due to catchment degradation, exacerbating hydrological vulnerability in the region as noted by the Central Water Commission. The 3.69% increase in barren land highlights soil erosion and localized resource extraction (such as quarrying), which strip the land of its natural ecological resilience.

2. Steps for Ecological Balance

• **Enforcement of Eco-Sensitive Zones (ESZs):** Strict implementation of zoning laws as recommended by the Kasturirangan Committee to restrict hazardous industrial and mining activities in vulnerable tracts.

• **Agroforestry Integration:** Transitioning from monoculture plantations to biodiversity-friendly agroforestry systems in buffer areas to restore soil fertility and create wildlife corridors.

• **Hydrological Restoration:** Rejuvenating degraded catchments through watershed management and afforestation to secure the water table and sustain regional hydrology.

3. Conclusive Points

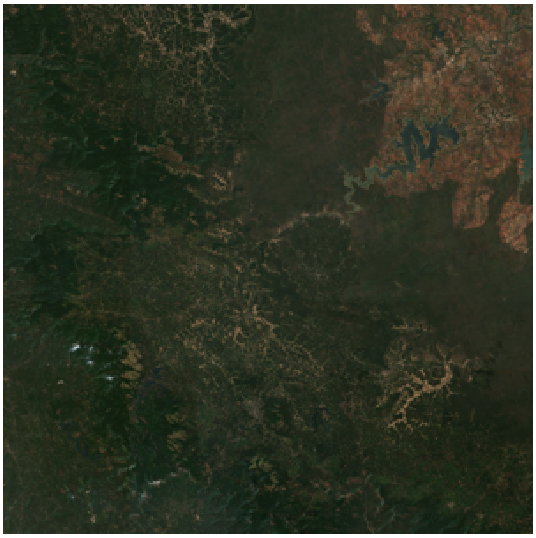
• **Habitat Fragmentation & Edge Effects:** The observed transitions in Wayanad reveal a concerning pattern. The conversion of natural habitats to anthropogenic land uses isolates endemic species and diminishes overall biodiversity capacity.

• **Unsustainable Urban Trajectory:** The direct replacement of ecological carbon sinks with impervious urban surfaces poses severe risks to localized climate resilience. It increases susceptibility to extreme weather events like landslides and flash floods, a recurring issue documented in recent regional climatic anomaly reports.

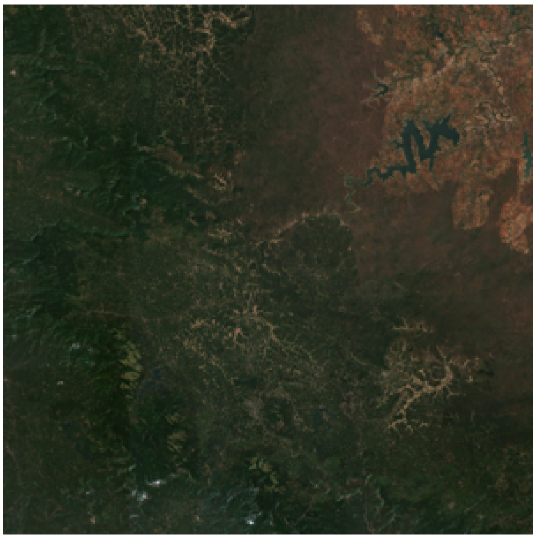
• **Imperative for Data-Driven Policy:** Real-time, satellite-derived insights emphasize the critical need for executing ground-level environmental regulations promptly. Continued unmonitored land-cover alterations will irreversibly compromise the long-term ecological and economic stability of peninsular India.

Visual Maps

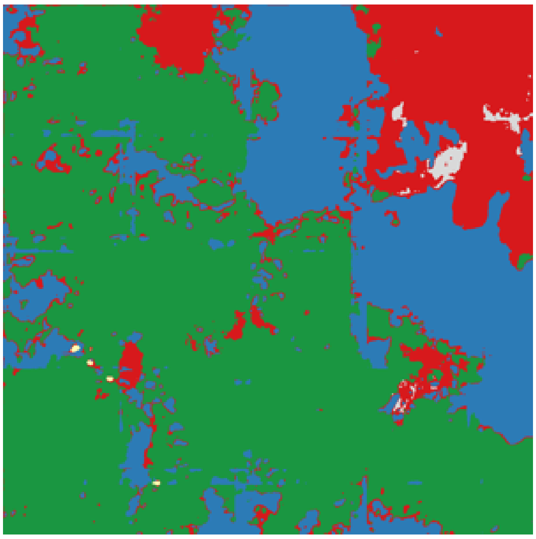
Satellite Image — Year 2025



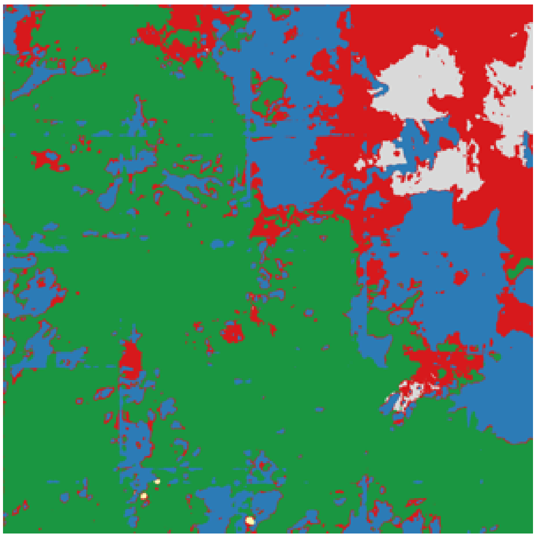
Satellite Image — Year 2026



Segmentation Map — Year 2025



Segmentation Map — Year 2026



Change Detection Map (Red=Forest Loss, Green=Forest Gain, Orange=Urbanisation)